1. INTRODUCTION

1.1 Project Overview

The global energy sector plays an important role in the economic growth and development of countries. Fuel is one of the most essential commodities because it is widely used in transportation, industries, agriculture, and everyday life. Changes in fuel prices directly affect the cost of living, transportation expenses, production costs, and the overall economy. Fuel taxation policies also differ from country to country, making it important to understand how these differences influence fuel prices across the world.

The project ***"Global Fuel Prices and Taxation Trends: A Comprehensive Analysis"*** focuses on analyzing global fuel prices and taxation trends from 2020 to 2026. The dataset covers 84 countries across 7 global regions and includes information about different fuel types, such as petrol, diesel, and liquefied petroleum gas (LPG). The project uses data analytics and visualization techniques to identify patterns, compare regional differences, and understand changes in fuel prices and taxation over time.

The project is developed using ***Tableau***, a data visualization tool that helps transform raw data into meaningful and interactive visual representations. Different visualization techniques, including geographic heat maps, trend charts, comparative bar charts, and regional performance indicators, are used to present the information clearly and effectively.

The analysis helps users explore variations in fuel prices, understand differences in taxation policies, and identify important trends across countries and regions. It also supports the interpretation of fuel-related economic and environmental factors.

Overall, this project demonstrates how data analytics and visualization can convert complex fuel-related data into understandable insights. These insights can be useful for policymakers, energy analysts, businesses, researchers, and other stakeholders interested in global fuel markets and taxation policies.



Figure 1.1: Global fuel market and fuel price analysis.

1.2 Purpose

Themain purposeofthis project is to analyze global fuel prices and taxation trends using data analytics and visualization techniques. It aims to present complex data in a simple and understandable format so that users can identify patterns, compare countries, and gain useful insights into the global fuel market. The major objectives of this project are as follows:

1. To Analyze Global Fuel Price Trends: To study changes in petrol, diesel, and LPG prices from 2020 to 2026 and understand how fuel prices vary over time.

2. To Compare Countries and Regions: To compare fuel prices and taxation patterns across 84 countries and 7 global regions and identify regional differences.

3. To Understand Taxation Policies: To explore the relationship between fuel prices and taxation structures and understand how tax policies may influence the final price of fuel.

4. To Create Interactive Visualizations: To use Tableau to develop charts, geographic heat maps, and comparative visualizations that make the data easier to explore and interpret.

5. To Support Data-Driven Decision-Making: To provide meaningful insights that can help policymakers, businesses, energy analysts, and researchers make informed decisions related to fuel pricing and taxation.

6. To Present Information Through a Dashboard and Story: To organize the visualizations into an interactive dashboard and a Tableau Story, allowing users to understand the analysis through a structured visual presentation.

In conclusion, this project aims to transform global fuel price and taxation data into meaningful visual insights. It helps users understand regional variations, observe historical trends, and explore the possible economic and environmental implications of fuel pricing policies.

2.IDEATION PHASE

2.1 problem statement

Date	26 Octomer 2026
Team ID	N.A.
Project Name	Global Fuel Prices and Taxation Trends
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

- ❑ Solve complex problems in a way that fits the state of your customers.
- ❑ Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior.
- ❑ Sharpen your communication and marketing strategy with the right triggers and messaging.
- ❑ Increase touch-points with your company by finding the right problem-behavior fit and
- ❑ building trust by solving frequent annoyances, or urgent or costly problems.

Understand the existing situation in order to improve it for your target group.

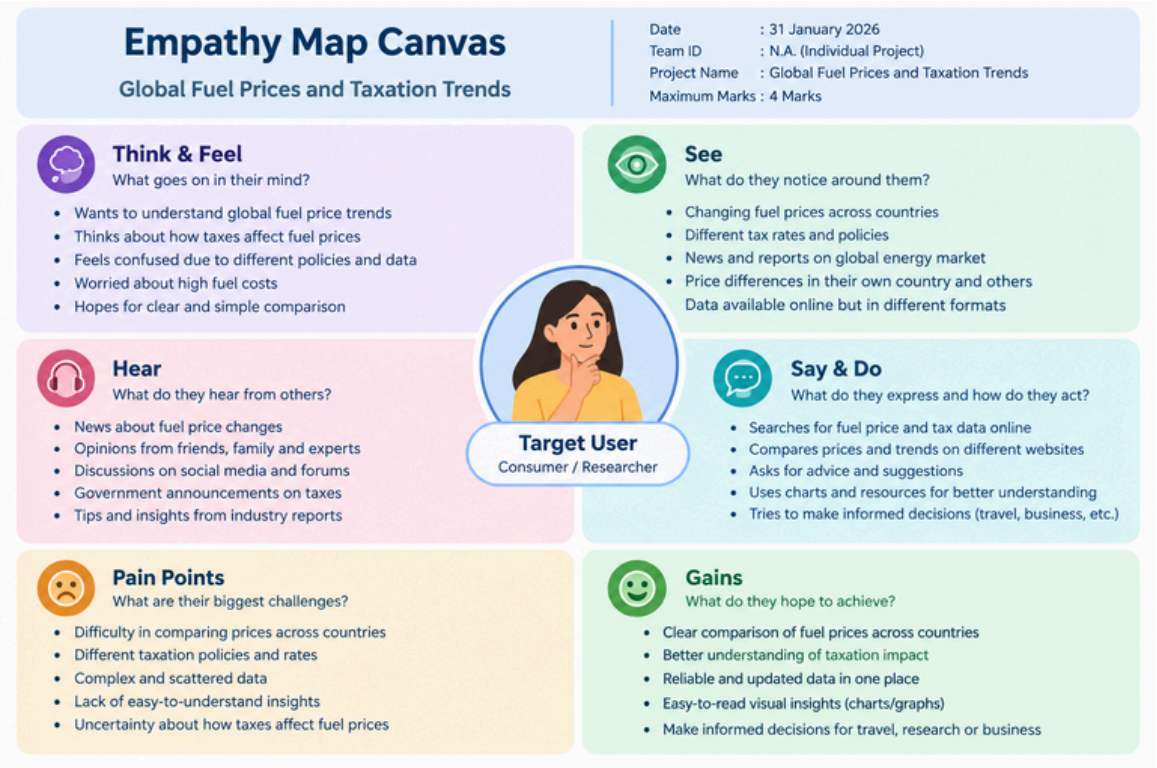


IdeationPhase
Empathize&Discover

Date	26 october 2026
Team ID	N.A.
Project Name	Global Fuel Prices and Taxation Trends
Maximum Marks	4Marks

Empathy Map Canvas:

This empathy map helps understand the needs, thoughts, and challenges of consumers and researchers who want to compare fuel prices and taxation across different countries. It identifies their problems, expectations, and the information they need to understand global fuel price trends.



IdeationPhase
Brainstorm & Idea Prioritization Template

Date	26 October 2026
Team ID	Individual Project
Project Name	Global Fuel Prices and Taxation Trends
Maximum Marks	4Marks

Step 1: Brainstorm and Idea Listing

The individual brainstorms different ideas related to the project without evaluating them at the initial stage.

S.No	Team Member	Idea / Suggestion	Category	Group No.
1	Kiran Jharbade	Analyze fuel price trends across countries	Price Analysis	N.A.
2	Kiran Jharbade	Compare fuel prices and taxation across countries	Taxation Analysis	N.A.
3	Kiran Jharbade	Study changes in fuel prices over time	Trend Analysis	N.A.
4	Kiran Jharbade	Compare fuel prices across different regions	Regional Analysis	N.A.

Step 2: Idea Prioritization

Evaluate each idea based on its feasibility and importance, then select the most suitable idea for the individual project.

Group No.	Final Idea	Feasibility (High/Medium/ Lo w)	Importance (High/Medium/ Lo w)	Priority	Selected (Yes/No)
1	Analyze global fuel price trends	High	High	1	Yes
2	Compare fuel prices and taxation	High	High	2	Yes
3	Study fuel price variations across regions	High	Medium	3	No

Customer Journey Map

Date	26 October 2026
Team ID	N.A.
Project Name	Global Fuel Prices and Taxation Trends
Maximum Marks	2 Marks

Customer Journey Map

Map out the customer's experience stage-by-stage, capturing their actions, touchpoints, thoughts, and feelings, along with process ownership and improvement opportunities at each stage.

Phase of Journey	**Stage 1 Identify**	**Stage 2 Analyze**	**Stage 3 Understand**
Actions *What does the customer do?*	Search fuel prices	Compare price data	Review trends
Touchpoint *What part of the service do they interact with?*	Websites, reports	Dataset, charts	Dashboard, graphs
Customer Thought *What is the customer thinking?*	Why do prices differ?	How do taxes affect prices?	What trends can I find?
Customer Feeling *What is the customer feeling?*	Confused	Curious	Informed

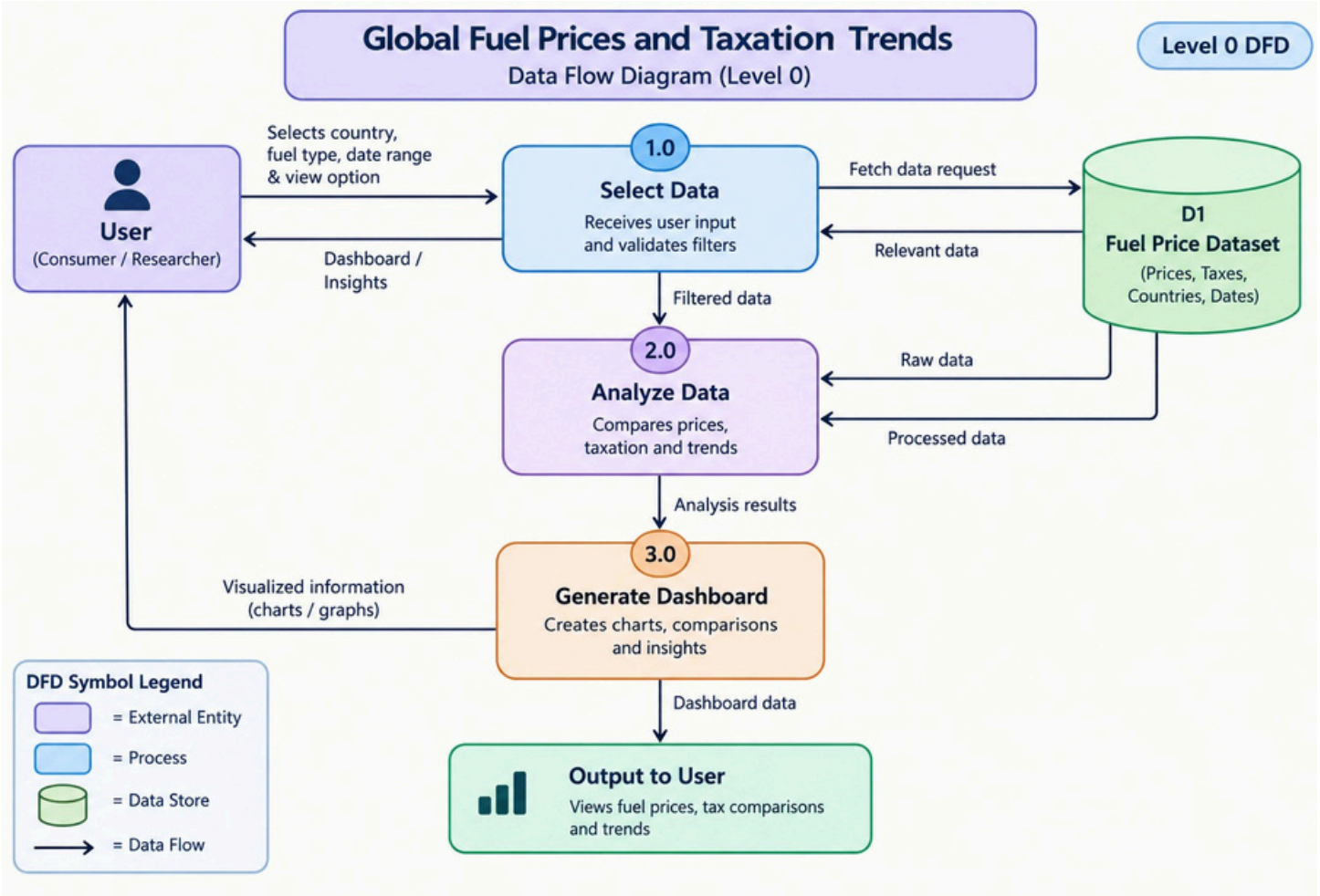
Project Design Phase-II
Data Flow Diagram & User Stories

Date	26 October 2026
Team ID	N.A.
Project Name	Global Fuel Prices and Taxation Trends
Maximum Marks	4 Marks

Data Flow Diagrams:
A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. A neat and clear DFD can depict the right amount of the system requirement graphically. It shows how data enters and leaves the system, what changes the information, and where data is stored.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	**External Entity**	A person, organization, or system outside the project that sends or receives data (e.g., Customer, Payment Gateway).
Rectangle with numbered header	**Process**	An activity that transforms incoming data into outgoing data (e.g., Validate Request, Process Order).
Rectangle (solid fill, no number)	**Data Store**	A place where data is held for later use (e.g., Customer Database, Inventory).
Labeled arrow	**Data Flow**	The movement of data between entities, processes, and data stores, labeled with what data is being passed.



User Stories

Usethebelowtemplate to list all the user stories for the product.

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Customer (Mobile user)	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	I can access my account / dashboard	High	Sprint-1
		USN-2	As a user, I will receive confirmation email once I have registered for the application	I can receive confirmation email & click confirm	High	Sprint-1
		USN-3	As a user, I can register for the application through Facebook	I can register & access the dashboard with Facebook Login	Low	Sprint-2
		USN-4	As a user, I can register for the application through Gmail		Medium	Sprint-1
	Login	USN-5	As a user, I can log into the application by entering email & password		High	Sprint-1
	Dashboard					
Customer (Web user)	N.A.	N.A.	No separate mobile application is included in this project.	Not applicable	N.A.	N.A.
Customer Care Executive	N.A.	N.A.	N.A.	N.A.	N.A.	N.A.
Administrator						
Administrator	Manage Dataset	USN-6	As an administrator, I want to update the fuel price dataset.	Updated data is available for analysis.	High	Sprint-1
Administrator	Check Data Quality	USN-7	As an administrator, I want to check the dataset for missing or incorrect values.	Data quality issues can be identified.	Medium	Sprint-2

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	26 October 2026
Team ID	N.A.
Project Name	Global Fuel Prices and Taxation Trends
Maximum Marks	2 Marks

Define Your Technology Stack:
Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1.	Programming / Query Language	Language used to query, filter, transform, aggregate, and analyze data.	Python, SQL.	Python helps analyze fuel price data and identify trends.
2.	Database / Data Warehouse	Warehouse Used to store and manage structured business data for analysis and reporting.	CSV Dataset	Stores fuel prices and related information in a structured format.
3.	Data Ingestion	Used to load data from source files or external systems into the analytical environment.	Snowflake Stages / COPY INTO / Tableau Data Connections	Imports the fuel price dataset for analysis.
4.	Data Transformation and Business Logic	Tools/Logic used to clean, join, filter, aggregate, data for analysis and calculate KPIs, metrics, business rules, and analytical measures.	Eg . Tableau prep, Snowflake SQL, Tableau Desktop	Helps clean data, handle missing values, and calculate summaries.
5.	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Snowflake SQl Charts, Tableau , Python	Visualizes fuel price comparisons and trends.
6.	Front-End / User Interface	Interface through which users interact with dashboards, filters, parameters, and visual insights.	Tableau Public, Flask, Streamlit	Provides a web interface to view charts, filters, and results

Project Design Phase-I
Problem – Solution Fit

Date	26 Octomer 2026
Team ID	N.A.
Project Name	Global Fuel Prices and Taxation Trends
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Purpose:

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- ❑ Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems.
- ❑ **Understand the existing situation in order to improve it for your target group.**



Project Initialization and Planning Phase

Date	26 october 2026
Team ID	N.A.
Project Title	Global Fuel Prices and Taxation Trends
Maximum Marks	3 Marks

Project Proposal (Proposed Solution)

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	Clearly state the project's primary objective
Scope	Define the boundaries and extent of the project
Problem Statement	
Description	Clearly articulate the problem to be addressed
Impact	Explain the implications of solving the problem
Proposed Solution	
Approach	Outline the methodology and techniques to be used
Key Features	Highlight the unique aspects of the proposed solution

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	e.g., 2 x NVIDIA V100 GPUs
Memory	RAM specifications	e.g., 8 GB
Storage	Disk space for data, models, and logs	e.g., 1 TB SSD

Software		
Frameworks	Python frameworks	e.g., Flask
Libraries	Additional libraries	e.g., scikit-learn, pandas, numpy
Development Environment	IDE, version control	e.g., Jupyter Notebook, Git
Data		
Data	Source, size, format	e.g., Kaggle dataset, 10,000 images

Project Design Phase-I
Solution Architecture

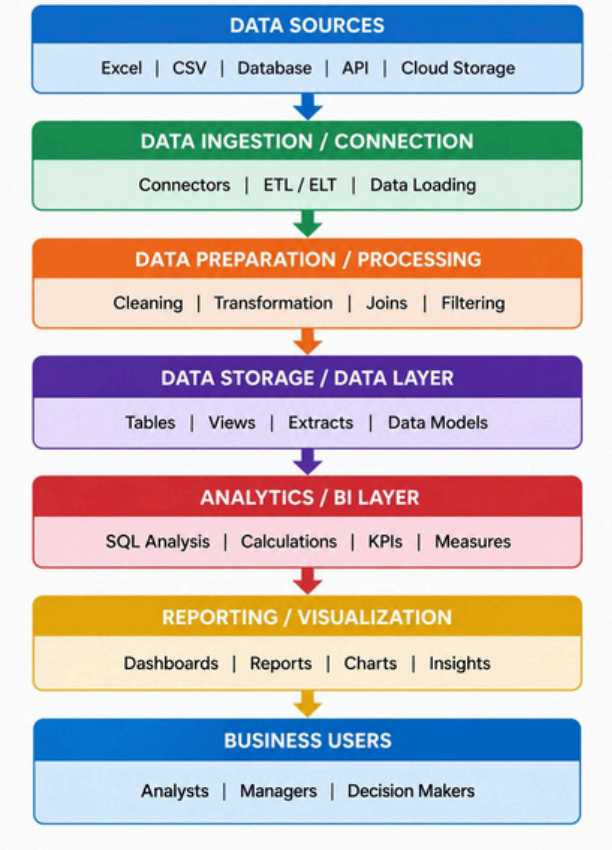
Date	26 october 2026
Team ID	N.A.
Project Name	Global flue price and texation
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Sample Solution Architecture Diagram:



Project Planning Phase

Project Planning Template(ProductBacklog,SprintPlanning, Stories, Story points)

Date	18 October 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Project - xxx
Maximum Marks	8 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

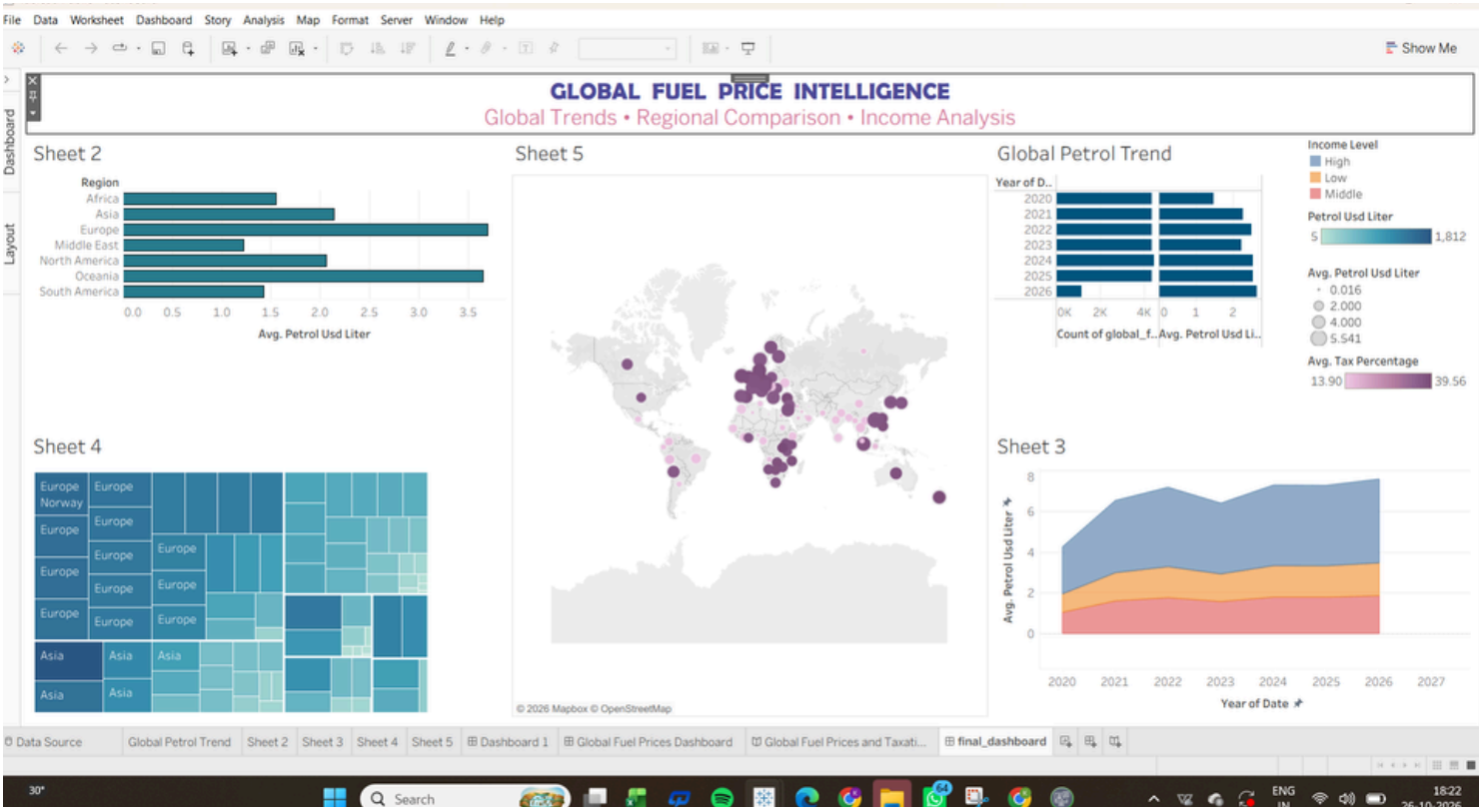
Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Registration	USN-1	As a user, I can register for the application by entering my email, password, and confirming my password.	2	High	Kiran jharbade
Sprint-1		USN-2	As a user, I will receive confirmation email once I have registered for the application	1	High	Kiran jharbade
Sprint-2		USN-3	As a user, I can register for the application through Facebook	2	Low	Kiran jharbade
Sprint-1		USN-4	As a user, I can register for the application through Gmail	2	Medium	Kiran jharbade
Sprint-1	Login	USN-5	As a user, I can log into the application by entering email & password	1	High	
	Dashboard					

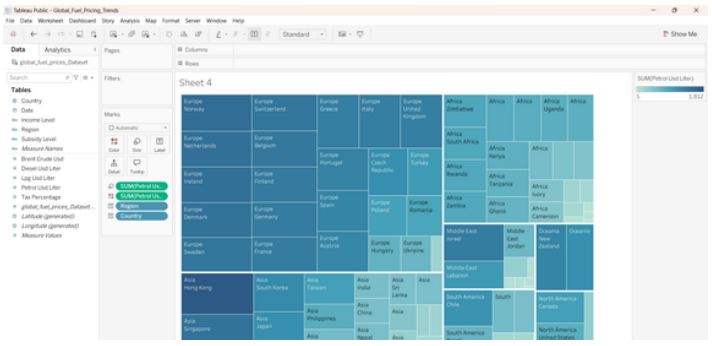
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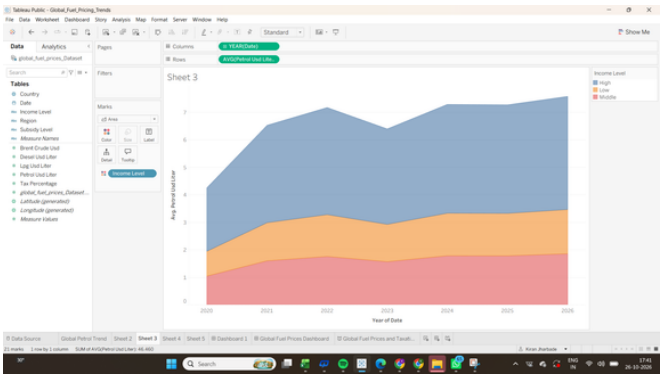
7. RESULTS

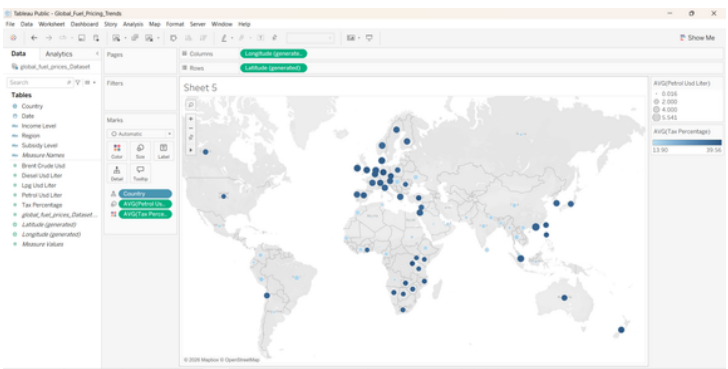
7.1 Output Screenshots

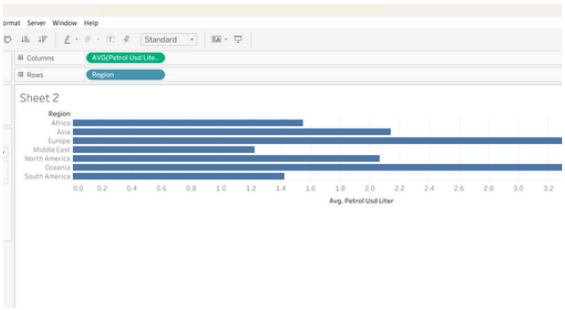


Dashboard









Visualizations

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

Global Fuel Prices and Taxation Trends: A Comprehensive Analysis

1. Easy Data Visualization

Tableau presents complex fuel price and taxation data through interactive charts, graphs, and maps, making it easier to understand.

2. Global Comparison

The project helps users compare fuel prices across 84 countries and 7 global regions to identify regional differences.

3. Trend Analysis

It allows users to explore changes in petrol, diesel, and LPG prices over the period from 2020 to 2026.

4. Better Decision-Making

The visual insights can help policymakers, energy analysts, and businesses understand fuel pricing patterns and support informed decisions.

5. Interactive Dashboard

The Tableau dashboard allows users to explore different visualizations and examine the data conveniently.

6. Understanding Taxation Patterns

The project helps users explore differences in fuel taxation across countries and investigate their relationship with fuel prices.

7. Time-Saving Analysis

Presenting important information through dashboards reduces the need to examine large amounts of raw data manually.

8. Useful for Multiple Users

The analysis can be useful for researchers, students, government organizations, businesses, and energy-sector professionals.

8.2 Disadvantages

Global Fuel Prices and Taxation Trends: A Comprehensive Analysis

1. Data Quality Dependency

The accuracy of the analysis depends on the quality, completeness, and reliability of the dataset.

2. Limited Country Coverage

Although the project covers 84 countries, it does not represent every country in the world.

3. Historical Data Limitations

The dataset covers the period from 2020 to 2026, so it may not explain long-term trends outside this period.

4. Complex Taxation Systems

Fuel taxation policies differ across countries, making direct comparisons difficult without considering local policies and economic conditions.

5. Limited Causal Analysis

Visualizations can show relationships and trends, but they cannot independently prove that taxation is the direct cause of fuel price changes.

6. Dependence on Tableau

Creating, editing, and exploring the visualizations depends on the availability of Tableau and familiarity with its features.

7. Data Updating Requirements

The dashboard may require regular dataset updates to reflect the latest fuel prices and taxation policies.

8. Visualization Limitations

Large datasets and multiple visualizations may sometimes make the dashboard difficult to interpret if it is not designed clearly.

9. CONCLUSION

The project “Global Fuel Prices and Taxation Trends: A Comprehensive Analysis” focuses on analyzing global fuel prices and taxation patterns using data analytics and visualization techniques. It covers data from 2020 to 2026 across 84 countries and 7 global regions, providing an opportunity to explore differences in fuel prices and taxation across the world.

Tableau is used to transform the dataset into meaningful visualizations, including geographic heat maps, trend charts, comparative bar charts, and regional performance indicators. These visualizations help present complex information in a clear and understandable manner. The dashboard and Tableau Story organize the analysis into a structured visual presentation, making it easier for users to explore the data and identify important patterns.

The project demonstrates the importance of data visualization in understanding changes in fuel prices, comparing regional differences, and exploring the relationship between fuel prices and taxation. It also highlights how interactive dashboards can make data analysis more accessible to students, researchers, businesses, and policymakers.

Although the analysis depends on the quality and coverage of the available dataset, it provides a useful foundation for exploring global fuel market trends. Overall, this project demonstrates how Tableau and data analytics can convert raw data into meaningful visual insights and support informed decision-making.

10. FUTURE SCOPE

The project “Global Fuel Prices and Taxation Trends: A Comprehensive Analysis” can be further improved by adding more data, advanced analytical techniques, and additional interactive features. The following enhancements can increase its usefulness in the future.

1. Expansion of Country Coverage

The dataset can be expanded to include more countries and regions to provide a broader understanding of global fuel prices and taxation policies.

2. Real-Time Data Integration

Real-time or regularly updated data sources can be integrated to help users monitor changes in fuel prices and taxation more frequently.

3. Fuel Price Prediction

Machine learning techniques can be explored to develop predictive models that estimate future fuel prices using historical data and relevant economic indicators.

4. Advanced Taxation Analysis

Further analysis can be performed to investigate how differences in fuel taxation relate to final fuel prices, government revenue, and regional economic conditions.

5. Environmental Impact Analysis

Future versions can include carbon emissions, fuel consumption, and renewable energy data to explore the relationship between fuel usage, taxation, and environmental sustainability.

11. APPENDIX

11.1 Dataset Link

The dataset used for this project contains information related to global fuel prices and taxation trends. It was used to analyze fuel price variations across different countries and regions from 2020 to 2026.

Dataset link : https://drive.google.com/file/d/13k0qQW8OqvyIrE3yCzAym5SUTmQNd0qu/view

11.2 GitHub Link

GitHub Link: https://github.com/kiranjharbade17/fuel-dataset

Tableau Public Dashboard Link:
https://public.tableau.com/app/profile/kiran.jharbade/viz/dashboard_17930048411860/final_dashboard

Tableau Public Story Link:
https://public.tableau.com/app/profile/kiran.jharbade/viz/Global_Fuel_Pricing_Trends/GlobalFuelPricesandTaxationStory

The dashboard and story were published on Tableau Public to make the visualizations accessible online. Users can explore the charts and understand global fuel price and taxation trends through the published visualizations.

11.3 Project Demo Link

Demo Video Link : https://drive.google.com/file/d/1-OIauZFg3WmXnbDTQEKJNv0j7zZ7V5ST/view?usp=drivesdk

A demo video has been recorded to explain the project, “Global Fuel Prices and Taxation Trends: A Comprehensive Analysis.” The video provides a step-by-step demonstration of the Tableau dashboard and story, including the visualizations, charts, maps, and key insights related to global fuel prices and taxation trends.